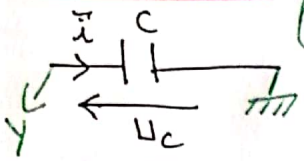
RC (Résumé)

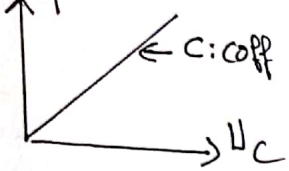
Condensatateur

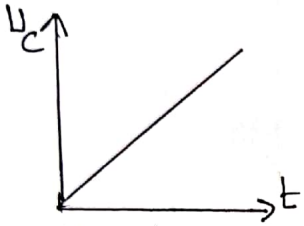


Branchement de l'oscilloscope pour mesurer U_C

$$q = C\,U_C$$

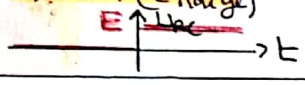
C : F, U_C : V



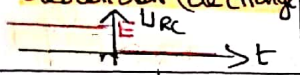


$U_C = K.t$

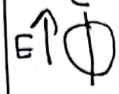
Echelon de tension montant (charge)



Echelon de tension descendant (décharge)

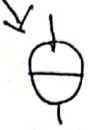


Intensité de courant



g. tension

$$i = C\frac{dU_C}{dt}$$

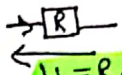


g. courant

$$I = \frac{q}{t}$$

Conducteur ohmique

Convention Recepteur



$U_R = R\,i$

Convention generateur

$U_R = -R\,i$

conditions initiales

Charge

* $U_C(0) = 0$
* $q(0) = 0$
* les autres $i(0) = \frac{E}{R_T}$; $U_R(0) = E$

Décharge

* $U_C(0) = U_C(\infty)_{charge}$
* $q(0) = q(\infty)_{charge}$
* $i(0) = -\frac{E}{R}$; $U_R(0) = -E$

Energie Emmagasinée

$$E_C(t) = \frac{1}{2}C U_C^2(t) = \frac{1}{2}\frac{q^2(t)}{C}$$
$$= \frac{1}{2}U_C(t).q(t).$$

Demonstration :

$$P_e = \frac{dE_e}{dt}$$
$$U_C.i(t) = \frac{dE_e}{dt}$$
$$C U_C \frac{dU_C}{dt} = \frac{dE_e}{dt}$$

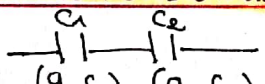
$$C\frac{d(\frac{1}{2}U_C^2(t))}{dt} = \frac{dE_e}{dt}$$

d'ou

$$E_e = \frac{1}{2}C U_C^2$$

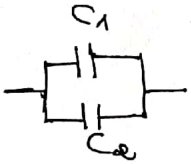
Association des condensateurs

* En Serie :



$$q_{eq} = q_1 = q_2$$
$$1/C_{eq} = 1/C_1 + 1/C_2$$

* En parallèle :



$$q_{eq} = q_1 + q_2$$
$$C_{eq} = C_1 + C_2$$

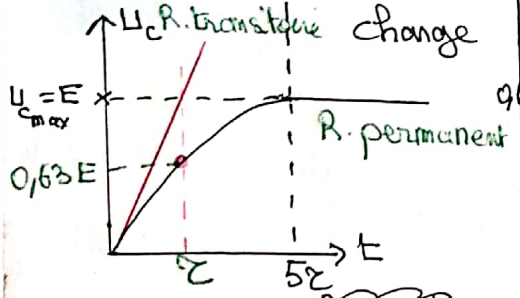
Dimension de τ

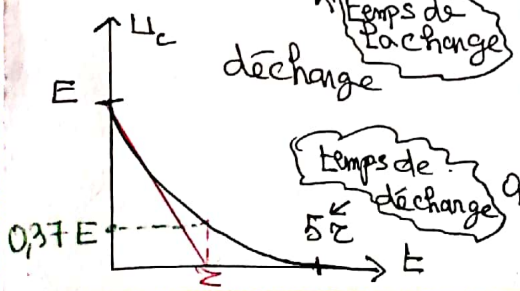
$$\tau = RC \Rightarrow [\tau] = [R].[C]$$
$$[R] = \frac{[U]}{[I]} \text{ et } i = C\frac{dU_C}{dt}$$
$$\Rightarrow [C] = \frac{[I].[T]}{[U]}$$

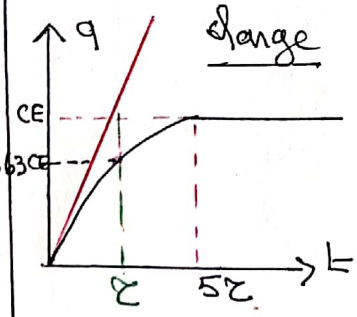
d'ou

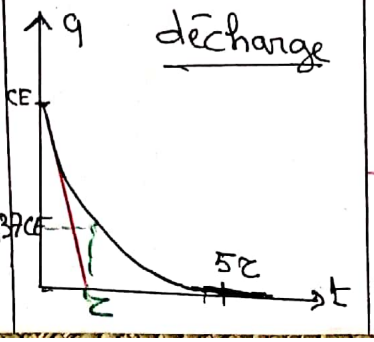
$$[\tau] = \frac{[U]}{[I]}.\frac{[I].[T]}{[U]}$$
$$[\tau] = [T]$$

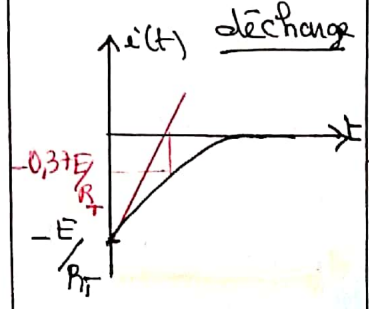
Comment determiner τ

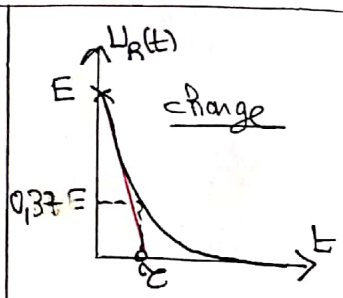


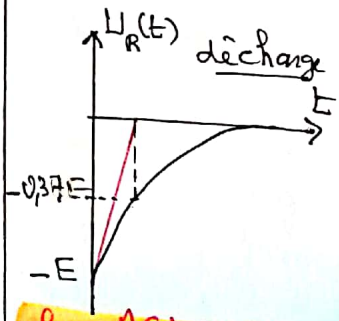












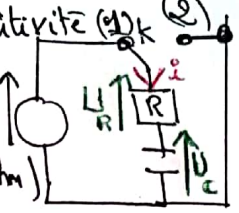
Pf: ASKOUR

Scanné avec CamScanner

Eq. diff en fct de $U_C(t)$ à la charge

* selon la loi d'additivité

$U_C(t) + U_R(t) = E$



* $U_R(t) = R\,i(t)$ (loi d'ohm)

$U_C(t) + R\,i(t) = E$

* $i(t) = C\dfrac{dU_C(t)}{dt}$

$$U_C + RC\frac{dU_C(t)}{dt} = E$$

$\Downarrow$

Eq. diff en fct de $q(t)$ à la charge

* selon la loi d'additivité

$U_C(t) + U_R(t) = E$

* loi d'ohm : $U_R(t) = R\,i(t)$

* $i = \dfrac{dq}{dt}$; $U_C(t) = \dfrac{q(t)}{C}$

$\dfrac{q(t)}{C} + R\dfrac{dq}{dt} = E$

$$q(t) + RC\frac{dq}{dt} = EC$$

$\Downarrow$

Eq. diff en fct de i à la charge

* la loi d'additivité

$U_C(t) + U_R(t) = E$

$U_C(t) + R\,i(t) = E$

$\dfrac{dU_C(t)}{dt} + R\dfrac{di(t)}{dt} = 0$

$\dfrac{i(t)}{C} + R\dfrac{di(t)}{dt} = 0$

$$i(t) + RC\frac{di(t)}{dt} = 0$$

$\Downarrow$

Solution de l'eq. diff. (Eq. horaire) (charge)

* $U_C(t) = Ae^{-\lambda t} + B$

* condition initial $U_C(0) = 0$

* $\dfrac{dU_C}{dt} = -\lambda A e^{-\lambda t}$

on remplace $U_C(t)$ et $\dfrac{dU_C(t)}{dt}$ dans l'Eq. diff.

$Ae^{-\lambda t} + B - RC\lambda A e^{-\lambda t} = E$

$Ae^{-\lambda t}(1 - RC\lambda) = E - B$

$\begin{cases} E - B = 0 \\ 1 - RC\lambda = 0 \end{cases} \Rightarrow \begin{cases} B = E \\ \lambda = 1/RC = 1/\tau \end{cases}$

* on a $U_C(0) = 0 \Rightarrow Ae^0 + B = 0$

$\Rightarrow A = -B = -E$

$$\Rightarrow U_C(t) = E(1 - e^{-t/\tau})$$

Solution de l'eq. diff (Eq. horaire)

$q(t) = Ae^{-\lambda t} + B$

$\Rightarrow$ condition initial : $q(0) = 0$

$\Rightarrow$ on remplace $q(t)$ et $\dfrac{dq(t)}{dt}$ dans l'eq. diff :

$Ae^{-\lambda t} + B - RC\lambda A e^{-\lambda t} = EC$

$Ae^{-\lambda t}(1 - RC\lambda) = EC - B$

$\begin{cases} EC - B = 0 \\ 1 - RC\lambda = 0 \end{cases} \Rightarrow \begin{cases} B = EC \\ \lambda = \dfrac{1}{RC} \end{cases}$

on a $q(0) = 0 \Rightarrow Ae^0 + B = 0$

$\Rightarrow A = -B = -EC$

$$\Rightarrow q(t) = CE(1 - e^{-t/\tau})$$

Solution de l'eq. diff (Eq. horaire)

$i(t) = Ae^{-\lambda t} + B$

$\Rightarrow$ condition initial

$i(0) = \dfrac{E}{R_T}$

on remplace $i(t)$ et $\dfrac{di(t)}{dt}$ dans l'eq. diff :

$Ae^{-\lambda t} + B - RC\lambda A e^{-\lambda t} = 0$

$Ae^{-\lambda t}(1 - RC\lambda) = -B$

$\begin{cases} -B = 0 \\ 1 - RC\lambda = 0 \end{cases} \Rightarrow \begin{cases} B = 0 \\ \lambda = \dfrac{1}{RC} \end{cases}$

$i(0) = \dfrac{E}{R_T} \Rightarrow Ae^0 = E/R$

$$\Rightarrow i(t) = \frac{E}{R}e^{-t/\tau}$$

décharge

K : l'interrupteur en position (2)

$U_C(t) + U_R(t) = 0$

$$\Rightarrow U_C(t) + RC\frac{dU_C(t)}{dt} = 0$$

* solution : $U_C(t) = Ee^{-t/\tau}$

condition initiale $U_C(0) = U_C(\infty)_{\text{charge}}$

$\Rightarrow U_C(0) = E$

décharge

$U_C(t) + U_R(t) = 0$

$$\Rightarrow q(t) + RC\frac{dq}{dt} = 0$$

Solution :

* $q(t) = CEe^{-t/\tau}$

condition initiale $q(0) = q(\infty)_{\text{charge}}$

$q(0) = CE$

décharge

$U_C(t) + U_R(t) = 0$

$$\Rightarrow i(t) + RC\frac{di}{dt} = 0$$

Solution :

$i(t) = -\dfrac{E}{R}e^{-t/\tau}$

condition initiale

$i(0) = -E/R$

Remarque : Astuce (calcul de l'instant t où le système (condensateur) à perdu $a\%$ de l'énergie maximale (càd à la fin de la charge)

$E(t) = (1 - a\%)E_{max}$

$\frac{1}{2}CU_C^2(t) = (1 - a\%)\frac{1}{2}CU_{Cmax}^2$

$\frac{1}{2}C(Ue^{-t/\tau})^2 = (1 - a\%)\frac{1}{2}CU_{Cmax}^2$

$\ln(e^{-t/\tau}) = \ln(\sqrt{1 - a\%})$

$-t/\tau = \ln\sqrt{(1 - a\%)}$

$t = -\tau\ln\sqrt{(1 - a\%)}$

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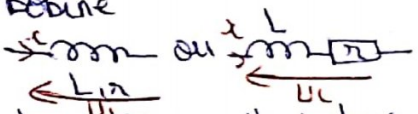
RL

bobine



L : coefficient d'inductance magnétique

r : résistance interne

$$U_L = L\frac{di}{dt} + ri$$

V, H, Ω, A

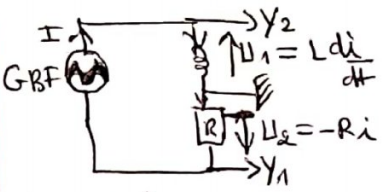
* le rôle de la bobine :

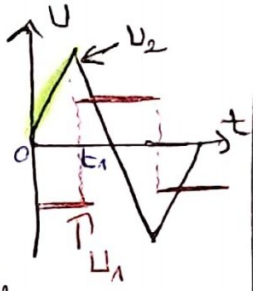
→ s'oppose à l'établissement et à la rupture du courant dans le régime transitoire.

→ dans le régime permanent : joue le rôle d'un conducteur ohmique de résistance r

* Rôle de la diode dans le circuit : impose le sens du courant et évite l'apparition de l'étincelle au niveau de l'interrupteur

* détermination expérimentale de L expérimentalement





$$U_1 = L\frac{di}{dt}$$

$$U_2 = -Ri \Rightarrow \frac{di}{dt} = -\frac{1}{R}\frac{dU_2}{dt}$$

$$U_1 = -\frac{L}{R}\cdot\frac{dU_2}{dt} \Rightarrow L = -R\frac{U_1}{\frac{dU_2}{dt}}$$

avec $\frac{dU_2}{dt}$: coeff directeur sur $[0, t_1]$

et U_1 valeur sur $[0, t_1]$

Energie magnétique stockée dans la bobine :

$$E_m(t) = \frac{1}{2}Li^2(t)$$

J, H, A

Conditions initiales

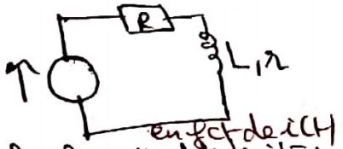
$i(0) = 0$; $i(t) = I_{max}$

Etablissement $E_{tab} = \frac{E}{R_T}$

$i(0)_{rupture} = \frac{E}{R_t}$

Eq. diff et solution (eq. Koraine)

Etablissement de courant :



en fct de i(t)

la loi d'additivité :

$$U_L(t) + U_R(t) = E$$

$$L\frac{di}{dt} + (r+R)i = E$$

$$i + \frac{L}{R+r}\frac{di}{dt} = \frac{E}{R+r}$$

$$i + \tau\cdot\frac{di}{dt} = \frac{E}{R+r}$$

$$\tau = \frac{L}{R+r}$$

en fct de $U_L(t)$ on determine l'eq diff en fct de i(t) puis on remplace i(t) par :

$$i(t) = \frac{E}{R} - \frac{U_L}{R}$$

et

$$\frac{di}{dt} = -\frac{1}{R}\frac{dU_L}{dt}$$

on trouve :

$$U_L + \frac{L}{R+r}\cdot\frac{dU_L}{dt} = \frac{Er}{R+r}$$

solution de l'eq. diff : $i(t) = Ae^{-\alpha t} + B$

$\frac{di}{dt} = -\alpha Ae^{-\alpha t}$ on remplace i(t) et $\frac{di}{dt}$ dans l'eq. diff :

$$-\frac{L}{R+r}\alpha Ae^{-\alpha t} + Ae^{-\alpha t} + B = \frac{E}{R+r}$$

$$Ae^{-\alpha t}\left(-\frac{L}{R+r}\alpha + 1\right) = \frac{E}{R+r} - B$$

cette relation n'est valable ∀t que si :

$$\begin{cases} \frac{E}{R+r} - B = 0 \\ -\frac{L}{R+r}\alpha + 1 = 0 \end{cases} \Rightarrow \begin{cases} B = \frac{E}{R+r} \\ \alpha = \frac{R+r}{L} = 1/\tau \end{cases}$$

et on a $i(0) = 0 \Rightarrow A + B = 0 \Rightarrow A = -B$

$A = -\frac{E}{R+r}$ d'où $i(t) = \frac{E}{R+r}(1 - e^{-t/\tau})$

Remarque :

L'étude de la rupture est similaire à celle de l'établissement sauf que :

$$U_L + U_R = 0$$

(loi d'additivité)

expression de $U_L(t)$:

$$U_L(t) = E - Ri(t)$$

$$U_L(t) = E - \frac{ER}{R+r}(1 - e^{-t/\tau})$$

dimension de τ :

$$\tau = \frac{L}{R_T} \; ; \; R_T = R + r$$

$$[R_T] = \frac{[U]}{[I]} \; ; \; U_L = L\frac{di}{dt}$$

$$[L] = \frac{[U].[T]}{[I]}$$

d'où $[\tau] = \frac{[L]}{[R]} = \frac{[U][T]}{[I]} \times \frac{[I]}{[U]}$

$$[\tau] = [T]$$

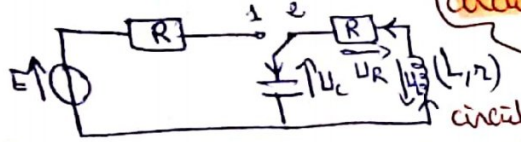
Scanné avec CamScanner

Scanné avec CamScanner

RLC

* Eq. diff en fct de $U_C(t)$:

circuit Réel $R_T \neq 0$



la loi d'additivité : $U_R(t)+U_L(t)+U_C(t)=0$

$$(R+r)i+L\frac{di}{dt}+U_C(t)=0$$

$$(R+r)C\frac{dU_C}{dt}+LC\frac{d^2U_C}{dt^2}+U_C(t)=0$$

$$\frac{d^2U_C(t)}{dt^2}+\frac{U_C(t)}{LC}+\frac{(R+r)}{L}\frac{dU_C}{dt}=0$$

* les régimes des oscillations :

R. périodique — T_0 : période propre

R. pseudo périodique — T : pseudo période, $T=T_0$

R. apériodique

Energie pour circuit Réel (RLC)

$\Delta E_T = E_T(t_2)-E_T(t_1)$, $t_1 \to t_2$

$T_e = \frac{T_0}{2}$ — période d'énergie

$E_T = E_e + E_m$

$$E_T=\frac{1}{2}CU_C^2+\frac{1}{2}Li^2(t)$$

$$\frac{dE_T}{dt}=CU_C\frac{dU_C}{dt}+Li\frac{di}{dt} \quad ; i=C\frac{dU_C}{dt}$$

$$\frac{dE_T}{dt}=CU_C\frac{dU_C}{dt}+LC^2\frac{dU_C}{dt}\cdot\frac{d^2U_C}{dt^2}$$

$$\frac{dE_T}{dt}=LC^2\frac{dU_C}{dt}\underbrace{\left(\frac{d^2U_C}{dt^2}+\frac{1}{LC}U_C\right)}_{-\frac{R+r}{L}\frac{dU_C}{dt}}=-(R+r)\left(C\frac{dU_C}{dt}\right)^2$$

← d'après l'eq. diff

$$\frac{dE_T}{dt}=-(R+r)i^2$$

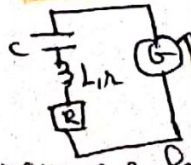
dissipation par effet Joule

$U_C+U_L+U_R=U_g$

$U_C+LC\frac{d^2U_C}{dt^2}+(r+R-K)i=0$

$$\frac{U_C}{LC}+\frac{d^2U_C}{dt^2}+\frac{(r+R-K)}{L}\frac{dU_C}{dt}=0$$

Entretient des oscillations :



$$\frac{U_C(t)}{LC}+\frac{d^2U_C}{dt^2}+\frac{r+R-K}{L}\frac{dU_C}{dt}=0$$

pour que le circuit devient comme idéal il faut que $r+R-K=0 \Rightarrow K=R+r$

circuit LC (idéal)

Eq diff :

$U_C+U_L=0$

$$U_C+LC\frac{d^2U_C}{dt^2}=0 \Rightarrow \frac{U_C}{LC}+\frac{d^2U_C}{dt^2}=0$$

* Eq en fct de q(t) : $\frac{q(t)}{LC}+\frac{d^2q}{dt^2}=0$

* solution de l'eq. diff :

$$U_C(t)=U_m\cos\left(\frac{2\pi}{T_0}t+\varphi\right)$$

U_m : valeur maximale de $U_C(t)$

T_0 : période propre

φ : phase à $t=0$

$\frac{2\pi}{T_0}=\omega_0$: pulsation propre.

* expression de T_0 :

$\frac{d^2U_C(t)}{dt^2}=-\left(\frac{2\pi}{T_0}\right)^2U_C(t)$; on remplace dans l'éq. diff

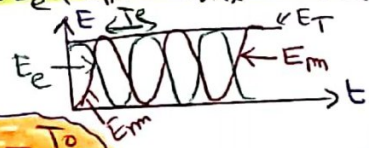
$$\frac{U_C}{LC}-\left(\frac{2\pi}{T_0}\right)^2U_C(t)=0 \Leftrightarrow U_C\left(\frac{1}{LC}-\left(\frac{2\pi}{T_0}\right)^2\right)=0$$

$$\left(\frac{2\pi}{T_0}\right)^2=\frac{1}{LC} \Rightarrow T_0=2\pi\sqrt{LC}$$

* φ est déterminée par $U_C(0)=U_m \Rightarrow \cos(\varphi)=1 \Rightarrow \varphi=0$

* Energie $E_T=$ cte pour un circuit LC

$$E_T=E_e+E_m=E_{e_{max}}=E_{m_{max}}$$



$T_e=\frac{T_0}{2}$

E.électrique — E.magnétique

* on démontre que $E_T=cte=E_e+E_m$

$$E_T=\frac{1}{2}CU_C^2+\frac{1}{2}Li^2 \Rightarrow \frac{dE_T}{dt}=CU_C\frac{dU_C}{dt}+Li\frac{di}{dt}$$

$$\Rightarrow \frac{dE_T}{dt}=CU_C\frac{dU_C}{dt}+LC^2\frac{dU_C}{dt}\cdot\frac{d^2U_C}{dt^2}$$

$$\Rightarrow \frac{dE_T}{dt}=LC^2\underbrace{\left(\frac{U_C}{LC}+\frac{d^2U_C}{dt^2}\right)}_{0}=0 \Rightarrow E_T=cste$$

* 2ème Méthode : $E_T=E_e+E_m=\frac{1}{2}CU_C^2(t)+\frac{1}{2}Li^2(t)$

et on a $U_C(t)=U_m\cos\left(\frac{2\pi}{T_0}t+\varphi\right)$ et $i(t)=-U_m\frac{2\pi}{T_0}\sin\left(\frac{2\pi}{T_0}t+\varphi\right)$

et $T_0=2\pi\sqrt{LC}$

d'où $E_T=\frac{1}{2}CU_m^2\cos^2\left(\frac{2\pi}{T_0}t+\varphi\right)+\frac{1}{2}LU_m^2C^2\frac{(2\pi)^2}{(2\pi\sqrt{LC})^2}\sin^2\left(\frac{2\pi}{T_0}t+\varphi\right)$

$$E_T=\frac{1}{2}CU_m^2\cos^2\left(\frac{2\pi}{T_0}t+\varphi\right)+\frac{1}{2}U_m^2C\sin^2\left(\frac{2\pi}{T_0}t+\varphi\right)$$

d'où $E_T=\frac{1}{2}CU_m^2\left[\cos^2\left(\frac{2\pi}{T_0}t+\varphi\right)+\sin^2\left(\frac{2\pi}{T_0}t+\varphi\right)\right]$

$$E_T=\frac{1}{2}CU_m^2$$

dimension de T_0 :

$[T_0]=[2\pi]\sqrt{[L][C]}$

$[T_0]=\sqrt{[L][C]}$

on a $U_L=L\frac{di}{dt}$

$[L]=\frac{[U].[T]}{[I]}$

et $i=C\frac{dU_C}{dt}$

$[C]=\frac{[I].[T]}{[U]}$

d'où $[T_0]=\sqrt{\frac{[U].[T]}{[I]}\cdot\frac{[I].[T]}{[U]}}$

$[T_0]=[T]$

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